

AN APPLICATION OF THE MOMENTUM EQUATION: DRAG OF A TWO-DIMENSIONAL BODY

$P=P_{\text{Inf}}$ on the CV $U_1=\text{Cte.}, U_2= F(y)$

Consider the surface forces on the control volume shown in Figure 2.18a. They stem from *two* contributions:

1. The pressure distribution over the surface *abhi*,

$$-\iint_{abhi} p \, dS$$

2. The surface force on *def* created by the presence of the body

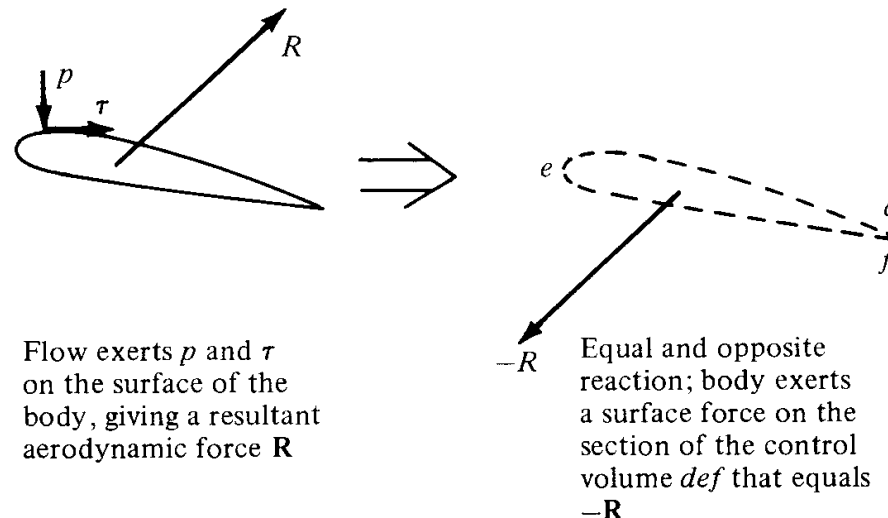


Figure 2.19

Equal and opposite reactions on a body and adjacent section of control surface.

AN APPLICATION OF THE MOMENTUM EQUATION: DRAG OF A TWO-DIMENSIONAL BODY

$$\text{Surface force} = - \iint_{abhi} p \, d\mathbf{S} - \mathbf{R}' \quad \mathbf{[2.73]}$$

Body force is negligible

Consider momentum Equation in integral form

$$\frac{\partial}{\partial t} \iiint_V \rho \mathbf{V} \, dV + \oiint_S (\rho \mathbf{V} \cdot d\mathbf{S}) \mathbf{V} = - \iint_{abhi} p \, d\mathbf{S} - \mathbf{R}' \quad \mathbf{[2.74]}$$

AN APPLICATION OF THE MOMENTUM EQUATION: DRAG OF A TWO-DIMENSIONAL BODY

Assuming steady flow, Equation (2.74) becomes

$$\mathbf{R}' = -\oint_S (\rho \mathbf{V} \cdot d\mathbf{S}) \mathbf{V} - \iint_{abhi} p d\mathbf{S} \quad [2.75]$$

Equation (2.75) is a vector equation. Consider again the control volume in Figure 2.18a. Take the x component of Equation (2.75), noting that the inflow and outflow velocities u_1 and u_2 are in the x direction and the x component of $\mathbf{R}'$ is the aerodynamic drag per unit span D' :

$$D' = -\oint_S (\rho \mathbf{V} \cdot d\mathbf{S}) u - \iint_{abhi} (p dS)_x \quad [2.76]$$

In Equation (2.76), the last term is the component of the pressure force in the x direction. [The expression $(p dS)_x$ is the x component of the pressure force exerted on the elemental area dS of the control surface.] Recall that the boundaries of the control volume $abhi$ are chosen far enough from the body such that p is constant along these boundaries. For a constant pressure.

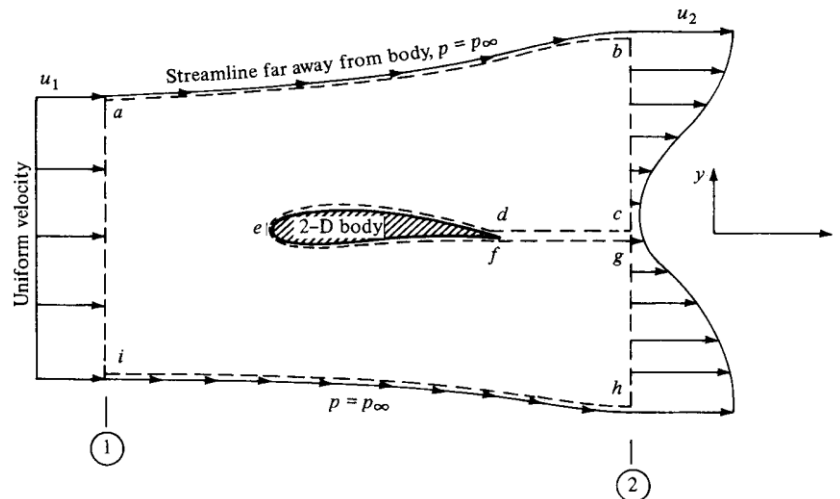
$$\iint_{abhi} (p dS)_x = 0 \quad [2.77]$$

$$D' = -\oint_S \rho \mathbf{V} \cdot d\mathbf{S} u$$

AN APPLICATION OF THE MOMENTUM EQUATION: DRAG OF A TWO-DIMENSIONAL BODY

Evaluating the surface integral in Equation (2.78), we note from Figure 2.18a that:

1. The sections ab , hi , and def are streamlines of the flow. Since by definition $\mathbf{V}$ is parallel to the streamlines and $d\mathbf{S}$ is perpendicular to the control surface, along these sections $\mathbf{V}$ and $d\mathbf{S}$ are perpendicular vectors, and hence $\mathbf{V} \cdot d\mathbf{S} = 0$. As a result, the contributions of ab , hi , and def to the integral in Equation (2.78) are zero.
2. The cuts cd and fg are adjacent to each other. The mass flux out of one is identically the mass flux into the other. Hence, the contributions of cd and fg to the integral in Equation (2.78) cancel each other.



AN APPLICATION OF THE MOMENTUM EQUATION: DRAG OF A TWO-DIMENSIONAL BODY

$dS = dy(1)$. The integral in Equation (2.78) becomes

$$\oint_S (\rho \mathbf{V} \cdot d\mathbf{S})u = - \int_i^a \rho_1 u_1^2 dy + \int_h^b \rho_2 u_2^2 dy \quad [2.79]$$

Note that the minus sign in front of the first term on the right-hand side of Equation (2.79) is due to $\mathbf{V}$ and $d\mathbf{S}$ being in opposite directions along ai (station 1 is an inflow boundary); in contrast, $\mathbf{V}$ and $d\mathbf{S}$ are in the same direction over hb (station 2 is an outflow boundary), and hence the second term has a positive sign.

Before going further with Equation (2.79), consider the integral form of the continuity equation for steady flow, Equation (2.53). Applied to the control volume in Figure 2.18a, Equation (2.53) becomes

$$- \int_i^a \rho_1 u_1 dy + \int_h^b \rho_2 u_2 dy = 0$$

or

$$\int_i^a \rho_1 u_1 dy = \int_h^b \rho_2 u_2 dy \quad [2.80]$$

Multiplying Equation (2.80) by u_1 , which is a constant, we obtain

$$\int_i^a \rho_1 u_1^2 dy = \int_h^b \rho_2 u_2 u_1 dy \quad [2.81]$$

Substituting Equation (2.81) into Equation (2.79), we have

$$\oint_S (\rho \mathbf{V} \cdot d\mathbf{S})u = - \int_h^b \rho_2 u_2 u_1 dy + \int_h^b \rho_2 u_2^2 dy$$

or

$$\oint_S (\rho \mathbf{V} \cdot d\mathbf{S})u = - \int_h^b \rho_2 u_2 (u_1 - u_2) dy \quad [2.82]$$

Substituting Equation (2.82) into Equation (2.78) yields

$$D' = \int_h^b \rho_2 u_2 (u_1 - u_2) dy$$

[2.83]